

- Q.30 Explain the effect of sound pollution due to aircrafts on humans.
- Q.31 Explain engine installation procedure.
- Q.32 What are the specifications of P&W PIGA engines?
- Q.33 What is the time line of different maintenance checks?
- Q.34 Explain the various controls in gas turbine engine
- Q.35 Explain the combustion process in the engine.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe Ant icing and cabin pressurization system. Explain the role and effect of an afterburner.
- Q.37 Explain in detail the various maintenance checks and inspections done on aircrafts.
- Q.38 What is the amount of pollutants released by aircrafts of different types? What is their effect on humans? Explain lubricating system.

No. of Printed Pages : 4
Roll No.

187762/147762

6th Sem / AME

Subject:- Turbo Propeller and Turbo Jet Engine - II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 How can regeneration be used to improve the efficiency of Brayton cycle?
- The energy of exhaust gas can be used to heat up the air which leaves the compressor
 - Heat supplied from external source thus decreases
 - the amount of heat rejected also decreases
 - all of the mentioned
- Q.2 The thermodynamic cycle for Gas Turbine is ____.
- Stirling Cycle
 - Brayton Cycle
 - Rankine cycle
 - Carnot cycle
- Q.3 The stage pressure ratio in axial flow compressor is about ____
- 1.7
 - 7
 - 10
 - 17

- Q.4 Thrust reversal on airliner is used to _____
- decrease the speed of the airplane
 - to increase the drag of the plane
 - To reverse the plane motion
 - To turn the plane
- Q.5 The air enters the combustion chamber in _____ parts
- One
 - Two
 - Three
 - Four
- Q.6 Thrust reversal system is used in airplanes for
- STOL
 - Landing
 - VTOL
 - all of the above
- Q.7 The ignition in gas turbine is initiated by?
- Pressure
 - Spark
 - Direct injection
 - Any of the above
- Q.8 The process in afterburner taking place is
- Compression of gases
 - Thrust addition by reheat
 - Turbine power enhancement
 - Heating after compressor
- Q.9 The type of aircraft gas turbines include
- Ram jet
 - Piston prop
 - turboprop
 - all of the above
- Q.10 What environmental effects can be due to aviation
- Sound pollution
 - Carbon oxides
 - O₃ depletion
 - All of the above

(2)

187762/147762

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Which fuel is used in gas turbines?
- Q.12 Why water injection is done in the engine?
- Q.13 Name lubricant used in Gas turbine engine?
- Q.14 What is the cabin pressure?
- Q.15 What is dressing of the engine?
- Q.16 What are the methods of fixing blades to the disc?
- Q.17 What are the main controls in engine?
- Q.18 What is special in PIGA engine?
- Q.19 Why hot section is done?
- Q.20 What is the use of guide vanes?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What type of pollution aircrafts create?
- Q.22 Name fuel system components and their function?
- Q.23 How turbine section is inspected?
- Q.24 How is dressing of gas turbine engine done?
- Q.25 How the igniter is tested?
- Q.26 How the engine thrust can be augmented?
- Q.27 Describe deicing procedures?
- Q.28 Explain the role of condition monitoring.
- Q.29 What is procedure for engine trimming.

(3)

187762/147762